

Spatial statistical modeling of shallow landslides—Validating predictions for different landslide inventories and rainfall events

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ARTICLE INFO

Article history:

Received 1 July 2010

Received in revised form 19 May 2011

Accepted 10 June 2011

Available online 17 June 2011

Keywords:

Landslide susceptibility

Logistic regression

Independent validation

Event-based landslide inventories

ABSTRACT

Statistical models that exploit the correlation between landslide occurrence and geomorphic properties are often used to map the spatial occurrence of shallow landslides triggered by heavy rainfalls. In many landslide susceptibility studies, the true predictive power of the statistical model remains unknown because the predictions are not validated with independent data from other events or areas. This study validates statistical susceptibility predictions with independent test data. The spatial incidence of landslides, triggered by an extreme rainfall in a study area, was modeled by logistic regression. The fitted model was then used to generate susceptibility maps for another three study areas, for which event-based landslide inventories were also available. All the study areas lie in the northern foothills of the Swiss Alps. The landslides had been triggered by heavy rainfall either in 2002 or 2005. The validation was designed such that the first validation study area shared the geomorphology and the second the triggering rainfall event with the calibration study area. For the third validation study area, both geomorphology and rainfall were different.

All explanatory variables were extracted for the logistic regression analysis from high-resolution digital elevation and surface models (2.5 m grid). The model fitted to the calibration data comprised four explanatory variables: (i) slope angle (effect of gravitational driving forces), (ii) vegetation type (grassland and forest; root reinforcement), (iii) planform curvature (convergent water flow paths), and (iv) contributing area (potential supply of water). The area under the Receiver Operating Characteristic (ROC) curve (*AUC*) was used to quantify the predictive performance of the logistic regression model. The *AUC* values were computed for the susceptibility maps of the three validation study areas (validation *AUC*), the fitted susceptibility map of the calibration study area (apparent *AUC*: 0.80) and another susceptibility map obtained for the calibration study area by 20-fold cross-validation (cross-validation *AUC*: 0.74). The *AUC* values of the first and second validation study areas (0.72 and 0.69, respectively) and the cross-validation *AUC* matched fairly well, and all *AUC* values were distinctly smaller than the apparent *AUC*. Based on the apparent *AUC* one would have clearly overrated the predictive performance for the first two validation areas. Rather surprisingly, the *AUC* value of the third validation study area (0.82) was larger than the apparent *AUC*. A large part of the third validation study area consists of gentle slopes, and the regression model correctly predicted that no landslides occur in the flat parts. This increased the predictive performance of the model considerably. The predicted susceptibility maps were further validated by summing the predicted susceptibilities for the entire validation areas and by comparing the sums with the observed number of landslides. The sums exceeded the observed counts for all the validation areas. Hence, the logistic regression model generally over-estimated the risk of landslide occurrence. Obviously, a predictive model that is based on static geomorphic properties alone cannot take a full account of the complex and time dependent processes in the subsurface. However, such a model is still capable of distinguishing zones highly or less prone to shallow landslides.

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1. Introduction

Shallow landslides have a significant impact on landscape evolution (Hovius et al., 1997; Korup et al., 2010) and are a risk for

infrastructure, threaten human life and cause loss of agricultural land. To prevent damage, models are required to predict zones prone to hazardous mass release. Two main classes of predictive models can be distinguished: Physically based (deterministic) and statistical (probabilistic) models. In the physically based approaches, a hydrologic model that describes subsurface water flow is coupled with an infinite slope stability analysis to evaluate the mechanical forces within the slope (Montgomery and Dietrich, 1994; Wu and Sidle, 1995). The

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predictive power of these approaches is usually strongly limited by scarce spatial information about the subsurface properties (e.g. soil depth, hydraulic conductivity, and shear strength) that control the spatio-temporal distribution of soil water and mechanical strength of the soil (Dietrich et al., 1995; Casadei et al., 2003; Guimarães et al., 2003; Godt et al., 2008).

The second class of predictive models exploits the correlation between landslide occurrence and explanatory variables such as slope angle, curvature, aspect, soil and bedrock properties. The basic assumptions of the statistical models are (i) that future landslides will occur under similar conditions as the observed ones (Guzzetti et al., 1999; Van Westen et al., 2008) and (ii) that properties deduced from digital elevation models (DEMs) are relevant for subsurface water flow and the related mechanical stability of the hillslopes. Unlike deterministic approaches, statistical methods require information on landslide occurrence (a landslide inventory) to estimate the coefficients of mathematical expressions that relate landslide susceptibility to the explanatory variables. Statistical models differ in how they parameterize these relations and how they estimate the coefficients. In the bivariate statistical approaches (e.g. Sørensen and Doyuran, 2004; Ayalew and Yamagishi, 2005; Song et al., 2008), which were proposed in several variants ('certainty factor', Chung and Fabbri, 1993; 'weights-of-evidence', Bonham-Carter et al., 1989; etc.), the observed values of each explanatory variable are grouped into a few classes (e.g. few groups of soil types or a set of intervals for slope angle), the frequency of landslide occurrence is estimated for each class and these estimates are merged for several explanatory variables. Discriminant analysis, a standard procedure to find linear or quadratic combinations of explanatory variables that separate two groups with maximum contrast, was also used to classify study-area units with and without landslides (Rossi et al., 2009). Another common approach is 'support vector machines' (nonlinear transformations of variables in a higher dimensional space) where the separation between units with and without landslides is maximized using various optimization and learning rules (Brenning, 2005). Artificial neural networks (Brenning, 2005; Ermini et al., 2005; Wang and Sassa, 2006; Falaschi et al., 2009) use a network of processing units (neurons). The communication between the units is determined by rules optimized to reproduce the output (landslide occurrence) as a function of explanatory variables.

In this study we employ logistic regression (Atkinson and Massari, 1998; Ohlmacher and Davis, 2003; Gorsevski et al., 2006; Nandi and Shakoor, 2009; Rossi et al., 2009) to model the spatial distribution of landslides triggered by heavy rainfall. A logistic regression model is a generalized linear model for a binary response variable (e.g. presence/absence of landslides). The transformed probability of occurrence, usually the 'logit' (logarithm of the probability ratio 'occurrence of landslide' and 'absence of landslide'), is modeled as a weighted sum of the explanatory variables. Sørensen and Doyuran, (2004), Ayalew and Yamagishi (2005) and Nandi and Shakoor (2009) found that logistic regression predicts susceptibility better than the bivariate methods, and Nefeslioglu et al. (2008), Yilmaz (2008) and Rossi et al. (2009) showed that its predictive performance compares well with artificial neural networks. Furthermore, Brenning (2005) showed that logistic regression is less prone to over-fitting the data than support vector machines.

No matter what statistical approach is used, one faces always the problem to validate the predictions derived from a statistical model. A common approach is to split the available data into a calibration subset for estimating the coefficients of the model, and into a validation subset for assessing its predictive performance. There are different ways to define the two subsets (see review of Chung and Fabbri, 2003): (i) The study area is split into two contiguous geographical areas with the calibration subset corresponding to 1/2 (Chung and Fabbri, 2008) or about 2/3 (Yesilnacar and Topal, 2005; Nefeslioglu et al., 2008; Nandi and Shakoor, 2009) of the study area.

(ii) A certain number of pixels (or grid cells) with landslides and an equal number of pixels without landslides (both randomly chosen) define the calibration subset and the remaining pixels of the study area are used for validation (García-Rodríguez et al., 2008; Nefeslioglu et al., 2008). (iii) Cross-validation: The data set is split into a small number (5–20) of subsets, and each subset is used in turn for validating the model fitted to the remaining subsets. This procedure is repeated for each subset and the performance measures, computed for each cross-validation subset, are averaged.

In all the studies cited above, the assessment of the predictive performance of the statistical model was limited by the fact that the model was calibrated and validated with data from the same study area, and it was therefore difficult to draw any general conclusions according to the predictive power for independent study areas. Statistical susceptibility predictions were hardly ever validated with independent data from other study areas. The studies by Lee (2005) and Domínguez-Cuesta et al. (2007) are notable exceptions: Both studies validated susceptibility predictions obtained by logistic regression for different validation areas and both found that the majority of the observed landslides lay in zones with large predicted susceptibility.

Our objective here is to test a logistic regression model rigorously by validating susceptibility predictions not only with data on landslides released during the same rainfall in another study area, but also with data on landslides triggered by a different heavy rainfall event. We had access to data of four event-based landslide inventories that had all been established by field surveys. Hence, each landslide could be attributed to a specific rainfall event. Two inventories referred to the same triggering event. We calibrated the logistic regression model with one of them and used the other inventory as a first validation data set. The study area of the landslide inventory used for calibration was close to another study area, for which landslides had been triggered in another year. The landslide inventory of the latter study area formed the second validation set. Because of spatial proximity, the calibration and the second validation study area were similar in terms of bedrock lithology, Quaternary history and, consequently the dominant landform shaping processes. In the sequel we use the terms similar (dissimilar) geomorphology when we refer to all factors jointly. Finally, for the third validation landslide inventory, both the geomorphology of the study area and the triggering rainfall were unique. With this choice of calibration and validation sets we were able to check transferability of landslide susceptibility predictions for different rainfall events and for different geomorphology.

2. Description of study areas, rainfall events and landslide inventories

We used four of the six landslide inventories compiled by Rickli and Graf (2009). The four inventories all refer to study areas in the northern foothills of the Swiss Alps, with Molasse as the dominant geological formation. Two inventories of Rickli and Graf (2009) were not considered because they are about landslides in study areas with different geology (Flysch and Helvetic nappes). The study areas and methods of landslide surveys are described in detail in Rickli et al. (2004), Raetzo and Rickli (2007) and Rickli and Graf (2009). In the remainder, the study areas are denoted by the name of the geographic region (Napf, Entlebuch or Appenzell) and the year when the landslides occurred (2002 or 2005).

2.1. Geology and geomorphology of study areas

As shown in Fig. 1a, the four study areas belong to two main tectonic units: Molasse Basin and Subalpine Molasse. In contrast to the Molasse Basin, the Subalpine Molasse was thrust northwards and uplifted in the course of the orogenesis of the Alps. Lithostratigraphically, the Molasse is divided into four formations: Upper Freshwater Molasse

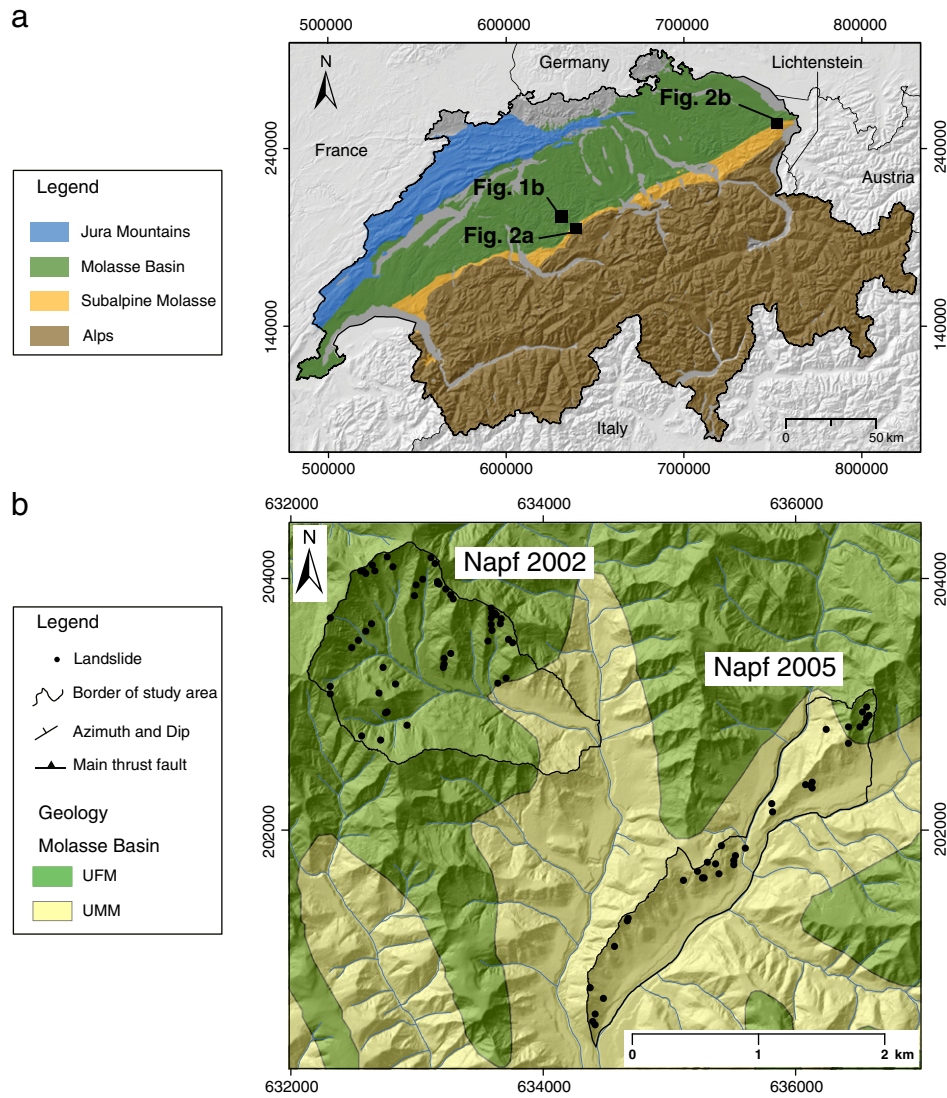


Fig. 1. Map of Switzerland with the main tectonic units and the location of the four study areas (a) and geologic map of the study areas in Napf region (b) with landslides triggered by rainfall events in 2002 and 2005 (black dots). The study area Napf 2002 is located mainly in the Upper Freshwater Molasse (UFM) and Napf 2005 in the Upper Marine Molasse (UMM).

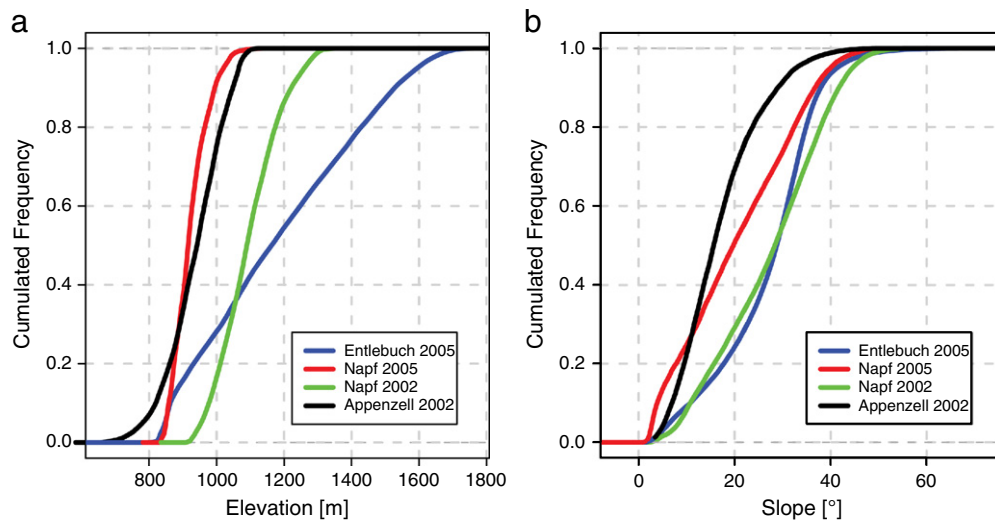


Fig. 2. Cumulative distribution functions of elevation (a) and slope angle (b) for the calibration study area Napf 2005 and for the validation areas Napf 2002, Appenzell 2002 and Entlebuch 2005. Compared to the other study areas, Entlebuch 2005 extends to higher altitude and covers a larger elevation range. Slope angles are approximately uniformly distributed for Napf 2005. In the case of Appenzell 2002 more than 70% of the study area has slope angles less than 20°, and steep slopes are found on only a small part of the study area.

(UFM, Langlian–Tortonian), Upper Marine Molasse (UMM, Burdigalian–Langlian), Lower Freshwater Molasse (LFM, Chattian–Aquitainian) and Lower Marine Molasse (LMM, Rupelian), in order of increasing age of the deposited material.

The study areas Napf 2002 and 2005 are part of the Molasse Basin (Fig. 1b). The bedrock is mostly UMM and UFM, dipping with 4–8° to northwest (Rickli et al., 2004; Rickli and Graf, 2009). The dominant geologic formation in Napf 2005 is UMM, consisting mainly of sands and marls, deposited in shallow marine and terrestrial conditions (Labhart, 2004). Napf 2002 is located almost exclusively in UFM, consisting of fluvial deposits of conglomerates and sandstones with intercalated marl layers. The entire Napf region was mostly ice-free

during the Last Glacial Maximum (Bini et al., 2009). Fluvial erosion and the associated hillslope processes (e.g. landslides) shaped the landscape and produced a dendritic fluvial network with relatively narrow valleys and steep hillslopes. For the Napf study areas, the ranges of elevation (Fig. 2a) are quite narrow (200–400 m), and the slope angles are approximately uniformly distributed (Fig. 2b).

The study area Entlebuch 2005 is part of the Subalpine Molasse (Fig. 3a) that was overthrust 15–25 km northwards and uplifted. The bedrock geology of Entlebuch 2005 is mainly LFM, dipping with 30°–40° to southeast (Labhart, 2004; Schwab et al., 2007). Similar to UFM in Napf 2002, the LFM layers in the Entlebuch area contain conglomerates and sandstones. The study area has wide valleys

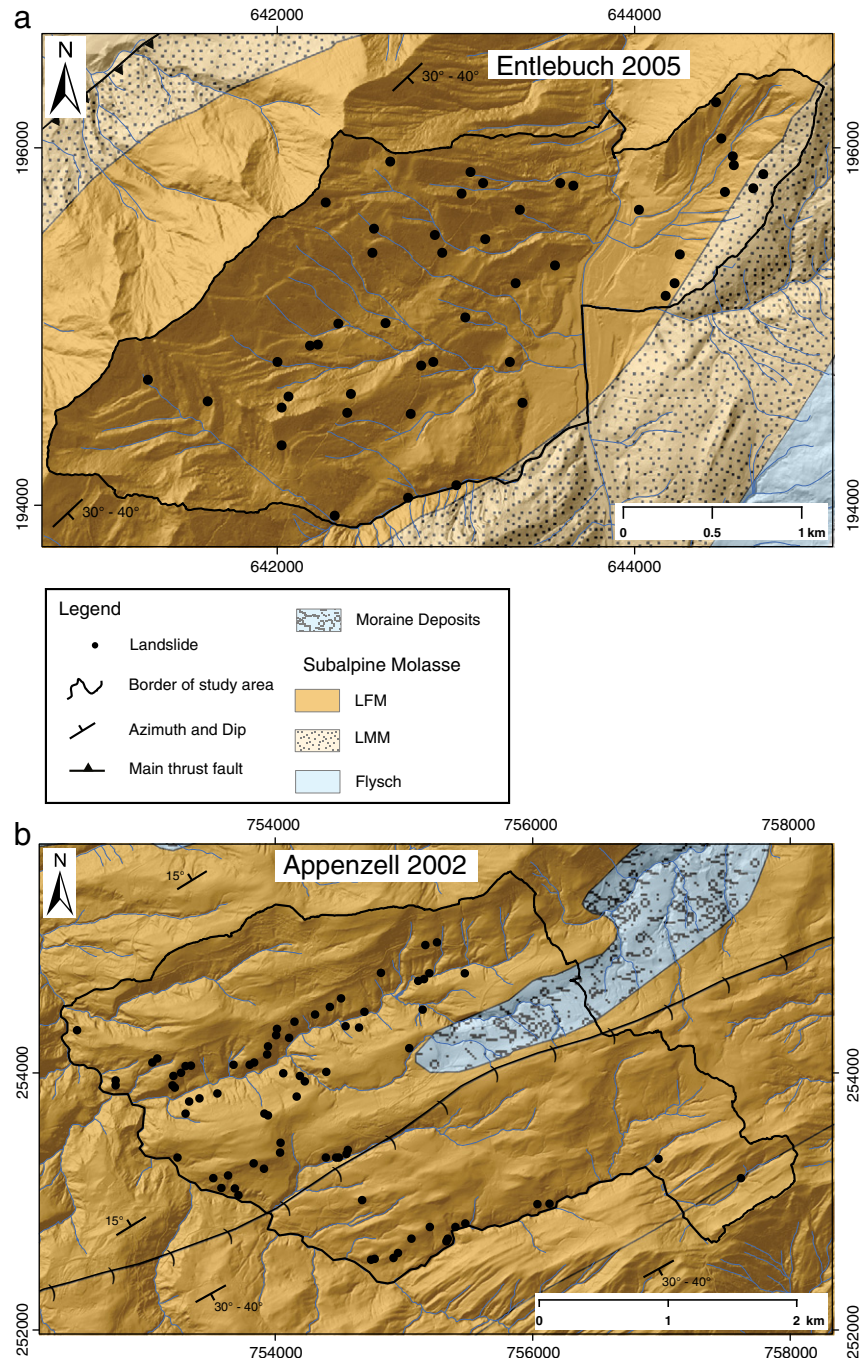


Fig. 3. Geologic map for study area Entlebuch (a) with landslides (black dots) triggered in 2005 (same rainfall event as for Napf 2005). The geological map for the study area Appenzell is shown in (b) with landslides (black dots) triggered by a rainfall event in 2002. Both study areas belong to the Lower Freshwater Molasse (LFM).

shaped by glacial erosion with moraines and fluvial deposits (Schlunegger, 2006; Schwab et al., 2007). The elevation ranges from 820 to 1680 m, and the distribution of the slope angles has a mode at about 30° (Fig. 2).

The Appenzell 2002 study area is also located in the Subalpine Molasse (Fig. 3b), and LFM prevails as well. While in the northern half of the study area the layers dip with 15° to north-northwest, layers in the south are inclined 30°–40° to southeast. The geomorphology of Appenzell 2002, with deep, narrow, V-shaped valleys and extended flat areas on plateaus, differs from the topography of the other study areas: slopes with angles less than 20° cover more than 70% of the total area.

2.2. Rainfall events

The first rainfall event considered in this study happened on 15 and 16 July, 2002. This local rainfall was short but very intense with 60 mm of rain in 3 h, inducing landslides in Napf 2002. According to Rickli et al. (2004), such an event has a recurrence period of 10 to 30 years. Another storm with 150 mm of rain in 12 h, in the night of 31 August to 1 September 2002, triggered many landslides in Appenzell. This event has a return period of 30–50 years (Rickli et al., 2004). The third rainfall event lasted from 18 to 23 August 2005 and triggered more than 5000 landslides across the northern part of the Swiss Alps including Napf 2005 and Entlebuch 2005 (Raetzo and Rickli, 2007). Rainfall amounts exceeded 100–150 mm in 48 h (on 21–22 August) and occasionally reached more than 200 mm. The return period for such an extreme hydrologic event is >100 years (Rotach et al., 2007). Table 1 summarizes some information about the rainfall events.

2.3. Landslide inventories

The Swiss Federal Institute for Forest, Snow and Landscape Research (WSL) conducted field surveys in the four study areas after the three rainfall events (two in 2002 and one in 2005) and compiled for each a detailed inventory of shallow landslides (further details in Rickli and Graf, 2009). In the field surveys, data were collected only for those landslides triggered by the respective rainfall. Based on 1:25,000 topographic maps the boundaries of the study areas were defined by local drainage divides and streams and were chosen under the constraint that the whole area could be visited and surveyed within a few days, resulting in surveyed areas <10 km². Note that for this study the boundaries were redefined to have closed hydrologic systems based on a 2.5×2.5 m digital elevation model; therefore, the area and portion of grassland presented in Table 1 differ from those reported by Rickli and Graf (2009). Only landslides with a minimum volume of 30 m³ and a depth (at the scar) of less than 2 m (i.e. shallow) were included in the inventory. Landslides occurring near streams were not analyzed to exclude the effect of fluvial undercutting by streams and rivers. In addition, landslides affected by roads or other man-made infrastructure were not considered because they were not controlled by effects related to surface and subsurface

topography. A total of 208 landslides with a median rupture surface of about 100 m² were recorded. In all cases not bedrock material but soil had been released. The observed mass release processes can be classified as rapid shallow landslides (Sidle and Ochiai, 2006) or earth slides (Varnes, 1978). About 85% of the landslides were translational (15% rotational). Apart from the position (measured by GPS), the length, width and depth of the landslides, topographic attributes (slope, aspect, and curvatures), soil type and hydrologic conditions were recorded in the surveys. Note that several attributes, determined only locally in field surveys, will not be included in the statistical model presented in this study, because they were not determined for the entire study area. The landslide positions are shown in Figs. 1 and 3 and some additional properties are listed in Table 1. The landslide density (number of landslides per km²) was much higher (>20 km⁻²) for the Napf than for the Appenzell and Entlebuch (<10 km⁻²) study areas.

3. Methods

We fitted a logistic regression model to the data of Napf 2005 because the landslides had been triggered by the same rainfall event as in Entlebuch 2005 and because the study area Napf 2002 was proximate. The study areas Napf 2002 and 2005 are not only spatially close (about 1 km apart) but also similar in terms of landscape forming processes, and both belong to the same tectonic unit (Molasse Basin; see Section 2.1 and Fig. 1a). By choosing Napf 2005 as calibration data set, we could test how well a statistical model predicts landslide susceptibility for geomorphologically 'similar' study areas but with different rainfall patterns; and for geomorphologically 'dissimilar' study areas but with landslides triggered by the same rainfall event. The study area Appenzell 2002 also served to validate the statistical model because the geomorphology and the rainfall event were completely different from Napf 2005.

Note that the objective of this study, namely to test transferability of statistical models affected the choice of analyzed data and explanatory variables. We considered only properties that could be transferred from one study area to another. For example, it is not meaningful to include geologic attributes like Upper Freshwater Molasse and Upper Marine Molasse as explanatory variable for the Napf area (see Fig. 1b) because they do not occur in the two other study areas. Below, the choice of explanatory variables is explained in detail.

3.1. Data preparation

Information on geology, soil type and land use was available only at poor spatial resolution (scale ≥1:200,000), resulting in only few map units per study area (see for example geologic maps in Figs. 1 and 3). Due to limited spatial resolution and the lack of transferability (see comment above) we ignored this information in the statistical analyses and extracted explanatory variables that we expected to affect landslide triggering mainly from elevation data (Table 2). Two digital high-resolution elevation data sets (cell size: 2.5×2.5 m)

Table 1
Characteristics of the three study areas (for more details see Rickli and Graf, 2009).

Study area	Area ¹ [km ²]	Grassland ¹ [%]	No. of slides [–]	Slide density [km ⁻²]	Rainfall amount ² [mm]	Rainfall duration [days]	Monthly rainfall ³ [mm]	Annual rainfall ³ [mm]
Napf 2002	2.4	45	51	21.3	60	1/8	213	1736
Napf 2005	1.4	64	35	25.0	236	4	170	1736
Entlebuch 2005	5.4	67	46	8.5	236	4	174	1669
Appenzell 2002	9.1	76	76	8.4	150	1/2	135	1386

¹ These values differ from the ones published by Rickli and Graf (2009) due to a redefinition of the boundaries of the study areas.

² Weather station for Napf and Entlebuch is located 5 km east and for Appenzell 7 km northeast of the respective study area.

³ Averaged monthly (annual) rainfall values for the corresponding month (year) when landslides occurred (values averaged over 30 years: 1961–1990).

Table 2
Explanatory variables used in the logistic regression analysis.

Explanatory variable	Expected relevance for landslide
(1) Slope angle [°]	Gravitational driving forces change with slope angle
(2) Vegetation type (forest, grassland)	Tree roots more effective to increase soil strength
(3) Profile curvature [m ⁻¹]	Acceleration/deceleration of water flow in downslope direction
(4) Planform curvature [m ⁻¹]	Convergent or divergent water flow affecting soil water content
(5) Common logarithm of contributing area [m ²]	Size of area routing water into a cell
(6) Distance to nearest drainage pathway [m]	Proxy for balance between in- and outflow of water into a cell

constructed from LIDAR data (Swisstopo, 2005) were available: one representing the elevation of the surface including vegetation and buildings, and the other about terrain elevation without them. Using GIS software we computed their difference, applied a threshold to the differences to determine whether the plant cover was forest or grassland and verified the vegetation classification with aerial images (1 × 1 m resolution). We calculated then the slope angle for each cell based on the largest height difference between the cell and its eight neighbors. The larger the slope angle, the higher the downslope driving forces. Driving forces are opposed by resisting normal and frictional stresses. Resisting forces are expected to increase with root reinforcement (Schwarz et al., 2010). Triggering conditions may differ between grassland and forest also with respect to other factors such as rainfall interception, evapotranspiration, and soil properties. For these reasons the vegetation type was used as an explanatory variable. The four remaining explanatory variables account for water flow and its effect on soil weight and mechanical strength: profile and planform curvatures control the water flow direction and were computed in a 3 × 3 moving window with the method of Zevenbergen and Thorne (1987). The contributing area, computed with the multiple flow direction algorithm of Tarboton (1997), describes accumulated inflow of water into a given cell. Finally, with the distance to the nearest flow paths, we included another measure of the balance between water in- and outflow in the statistical analysis.

In the first step all explanatory variables were computed for a 2.5 × 2.5 m grid. We assumed that a landslide results from progressive failure (e.g. Petley et al., 2005): the mass release is initiated somewhere on the rupture surface of the landslide and not necessarily at the scar. For this uncertainty we reduced the grid resolution to 10 × 10 m using the median value of 4 × 4 cells of the original grid, considering the median area of the rupture surfaces (~100 m²). The categorical variable 'vegetation type' was aggregated by assigning the dominating vegetation type in the 16 original cells to a 10 × 10 m cell.

3.2. Logistic regression analysis

The spatial incidence of shallow landslides was modeled by logistic regression, which is a well-established method to model binary or binomial data (e.g. Hosmer and Lemeshow, 2000). Logistic regression relates the probability, π_i , of a binary event (landslide occurs in i -th grid cell) to a set of p explanatory variables, x_{ik} , $k = 1, 2, \dots, p$, by

$$g(\pi_i(x_{i1}, x_{i2}, \dots, x_{ip})) = \mu_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}, \quad (1)$$

where $g(\pi_i)$ is the link function of π_i , μ_i is the linear predictor, and β_k are the regression coefficients. We used the logit link function

$$g(\pi_i) = \mu_i = \log\left(\frac{\pi_i}{1 - \pi_i}\right), \quad (2)$$

to relate μ_i to the odds(π_i) = $\pi_i/(1 - \pi_i)$ of the event.

The regression coefficients convey information how the odds ratios change with a unit change of the respective explanatory variable when the other explanatory variables are held constant (Hosmer and Lemeshow, 2000). For a binary explanatory variable (vegetation type in our study), $x_{ik} = 1$ if a particular condition holds for the i -th observation (= i -th grid cell is grassland), and $x_{ik} = 0$ otherwise. The coefficient β_k is equal to

$$\beta_k = \log\left(\frac{\text{odds}(\pi_i(x_{i1} = c_1, \dots, x_{ik} = 1, \dots, x_{ip} = c_p))}{\text{odds}(\pi_i(x_{i1} = c_1, \dots, x_{ik} = 0, \dots, x_{ip} = c_p))}\right), \quad (3)$$

where c_j is a constant. If π_i is small (<5%) then $\text{odds}(\pi_i) \approx \pi_i$ and therefore

$$\beta_k \approx \log\left(\frac{\pi_i(x_{i1} = c_1, \dots, x_{ik} = 1, \dots, x_{ip} = c_p)}{\pi_i(x_{i1} = c_1, \dots, x_{ik} = 0, \dots, x_{ip} = c_p)}\right). \quad (4)$$

For rare events, $\exp(\beta_k)$ is equal to the relative risk that is associated with the particular condition (Hosmer and Lemeshow, 2000):

$$\exp(\beta_k) \approx \frac{\pi_i(x_{i1} = c_1, \dots, x_{ik} = 1, \dots, x_{ip} = c_p)}{\pi_i(x_{i1} = c_1, \dots, x_{ik} = 0, \dots, x_{ip} = c_p)}. \quad (5)$$

If x_k is a continuous explanatory variable (e.g. slope angle), the exponential of β is equal to the relative risk that would result if x_k were increased by one unit and the other explanatory variables would remain unchanged.

The β coefficients were estimated based on maximum likelihood (Hosmer and Lemeshow, 2000) for the calibration study area Napf 2005 using the R software package (R Development Core Team, 2011). The full data set (35 cells with landslides and 14,140 without them) was used to fit the model. The final model was determined using stepwise backward variable selection: starting from the 'full' model that included all six explanatory variables, the set of explanatory variables was step-by-step reduced (and possibly re-expanded) to minimize the Akaike Information Criterion (AIC).

The quality of the model was examined by goodness-of-fit tests and customary residual diagnostic plots (Hosmer and Lemeshow, 2000). The plots indicated that the contributing area should be transformed to the common logarithm, and the curvature variables had to be winsorized to reduce the influence of single observations or outliers. Therefore all values outside the 1st and 99th percentile were trimmed.

3.3. Receiver operating characteristics

The predictive performance of the final model was assessed by Receiver Operating Characteristics (ROC) curves (e.g. Fawcett, 2006), which measure the discriminating power or the conditional bias of a classification method (Wilks, 2006). An ROC curve displays, for a sequence of monotonically increasing probability thresholds π_t in [0,1], pairs $[FP(\pi_t), TP(\pi_t)]$ in a line chart, where $FP(\pi_t)$ is the false positive rate (percentage of misclassified landslide-free cells; 1-specificity) and $TP(\pi_t)$ is the true positive rate (percentage of correctly classified cells with landslides; sensitivity). To compute these rates for a threshold π_t , the predicted probabilities π_i were converted to binary numbers using the indicator

$$I(\pi_i; \pi_t) = \begin{cases} 1 & \text{if } \pi_i > \pi_t \\ 0 & \text{if } \pi_i \leq \pi_t \end{cases}. \quad (6)$$

Note that $I(\pi_i; \pi_t) = 1$ means that a landslide is predicted for the i -th grid cell. The area under the ROC curve (AUC) is a measure of the

Table 3

Coefficients of explanatory variables of the final logistic regression model with standard errors, z-values (ratio of estimate and standard error) and nominal p-values. The coefficient β_3 describes by how much the intercept of the regression model differed for sites on grassland relative to sites in forest (β_0 is the intercept for forest and $\beta_0 + \beta_3$ for grassland sites, see text for further interpretation of the coefficients).

Coefficients β_k	Estimate	Std. Error	z-value	p-value ^a
Intercept for forest β_0	−11.86	1.48	−	−
Slope β_1	0.12	0.02	5.43	5.4×10^{-8}
Planform curvature β_2	−17.93	6.74	−1.92	5.5×10^{-2}
Effect of grassland β_3	1.68	0.45	3.68	2.2×10^{-4}
Log ₁₀ [contributing area] β_4	0.77	0.47	1.63	0.1

^a Based on z-test.

conditional bias of a binary classification method. $AUC = 0.5$ results if a method is unable to discriminate, and $AUC = 1$ shows perfect discrimination. Hosmer and Lemeshow (2000) consider an AUC value of 0.7 as acceptable and 0.8 as excellent discrimination.

ROC curves were generated for both the validation and calibration study areas. For the latter, we obtained susceptibilities by fitting the model to the Napf 2005 calibration data. It is well known that the

apparent ROC curve tends to be too optimistic because the same data is used for calibration and testing (e.g. Hastie et al., 2009). A common strategy against over-optimism is cross-validation (e.g. Hastie et al., 2009), splitting the study areas in subregions. Brenning (2005) showed that cross-validation with randomly chosen, non-contiguous subsets tends to overestimate the predictive performance. Therefore, the calibration study area Napf 2005 was split into 20 spatially contiguous sub-regions and each sub-region was used once to validate the model fitted to the data of the remaining 19 sub-regions. Stepwise selection of the explanatory variables was repeated for each set of sub-regions.

4. Results

4.1. Calibrating the logistic regression model with data of the study area Napf 2005

The stepwise procedure for selecting the explanatory variables stopped with a final regression model consisting of the four explanatory variables: slope, planform curvature, vegetation type and contributing area. The estimated regression coefficients and their

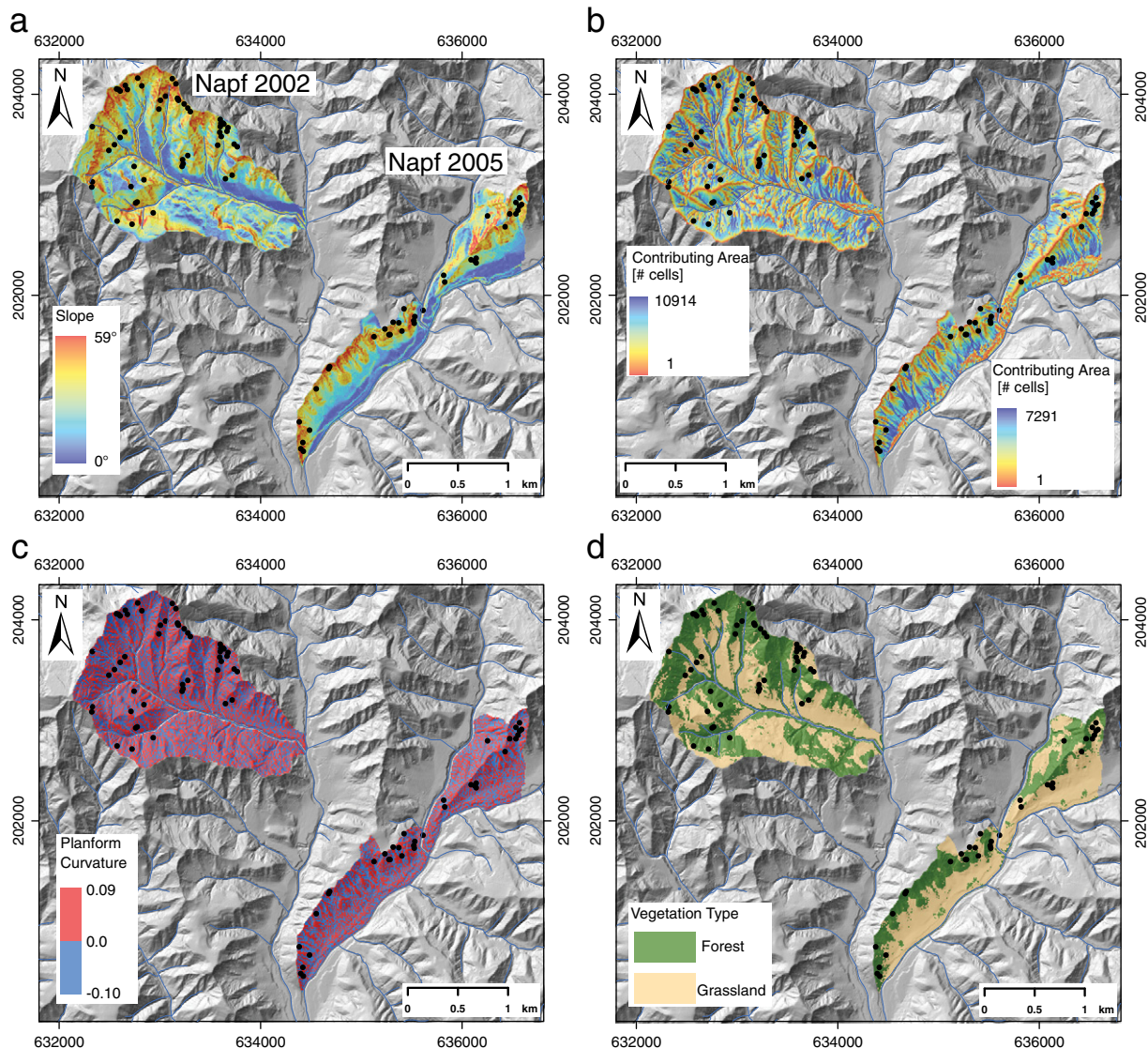


Fig. 4. Maps of explanatory variables (10 m grid spacing) used to model landslide occurrence for study areas Napf 2002 and 2005 with locations of shallow landslides (black dots): (a) slope angle, (b) contributing area, (c) planform curvature, and (d) vegetation type. The study area Napf 2005 was used for model calibration.

standard errors are listed along with the p -values in Table 3. Figs. 4–6 show the spatial distributions of the explanatory variables in the four study areas. While slope, vegetation type and planform curvature were significant (p -values < 5%), the p -value of contributing area was 10%. However, eliminating this variable increased AIC distinctly, and therefore we kept it in the model.

The fitted susceptibilities π_i were small (median 0.0013; $\pi_i > 0.05$ only for a few cells); hence the approximation of Eq. (4) could be used for the interpretation of the regression coefficients. The coefficient for slope angle (β_1 in Table 3) was 0.12. Increasing the angle by 10° corresponds to a multiplication of the susceptibility by a factor of $\exp(1.2) = 3.3$. The regression coefficient for planform curvature β_2 was -17.93 , indicating higher landslide risks for concave slopes. Making the curvature more pronounced by decreasing the planform curvature for example by one standard deviation ($= 0.02$) leads to change in the relative risk of landslides by a factor of $\exp(17.93/50) = 1.43$. The coefficient β_3 for the vegetation type was 1.68. By changing forest to grassland, the risk of landslides increases by a factor of $\exp(1.68) = 5.4$. Finally, the coefficient of the common logarithm of contributing area β_4 was 0.77. Doubling the contributing area of a cell increases the landslide susceptibility by $2^{0.77/\log 10} = 1.26$.

4.2. Validating the model for Napf 2002, Appenzell 2002 and Entlebuch 2005

The landslide susceptibility maps for the calibration and validation study areas (Figs. 7 and 8) show that the landslides occurred mostly in zones where the predicted susceptibility is high. However, not all

zones with elevated susceptibility experienced many landslides: In particular for Entlebuch 2005 (Fig. 8a) and, to a lesser extent for Napf 2002 (Fig. 7a), most zones with $\pi_i > 0.01$ (reddish colors) were free of landslides, suggesting that the final model over-predicted landslide risk, at least for Entlebuch 2005.

We tested the above inference by summing the predicted π_i values for all grid cells and by comparing it with the observed counts of landslides. Note that the sum of π_i is equal to the observed number of landslides for the calibration study area. For the validation study areas, the sums were larger than the observed counts: their ratio was about 2 for Napf 2002 and Appenzell 2002 and 7 for Entlebuch 2005. Hence, the predictions were indeed positively biased.

Fig. 9 shows the ROC curves for the three validation study areas with the apparent and the cross-validation ROC curves for Napf 2005. The AUC values range from 0.69 to 0.82, and the largest is found for the validation curve Appenzell 2002. A large fraction of Appenzell 2002 is flat, and the regression model correctly predicted small π_i for the flat parts. As expected, the apparent curve is too optimistic (AUC value 0.80) compared with the cross-validation curve for Napf 2005 (AUC = 0.74) and the validation curves for Napf 2002 (0.72) and Entlebuch 2005 (0.69).

5. Discussion

5.1. Choice of explanatory variables

The coefficients of the model indicate that landslide susceptibility increases with increasing slope angle, increasing contributing area,

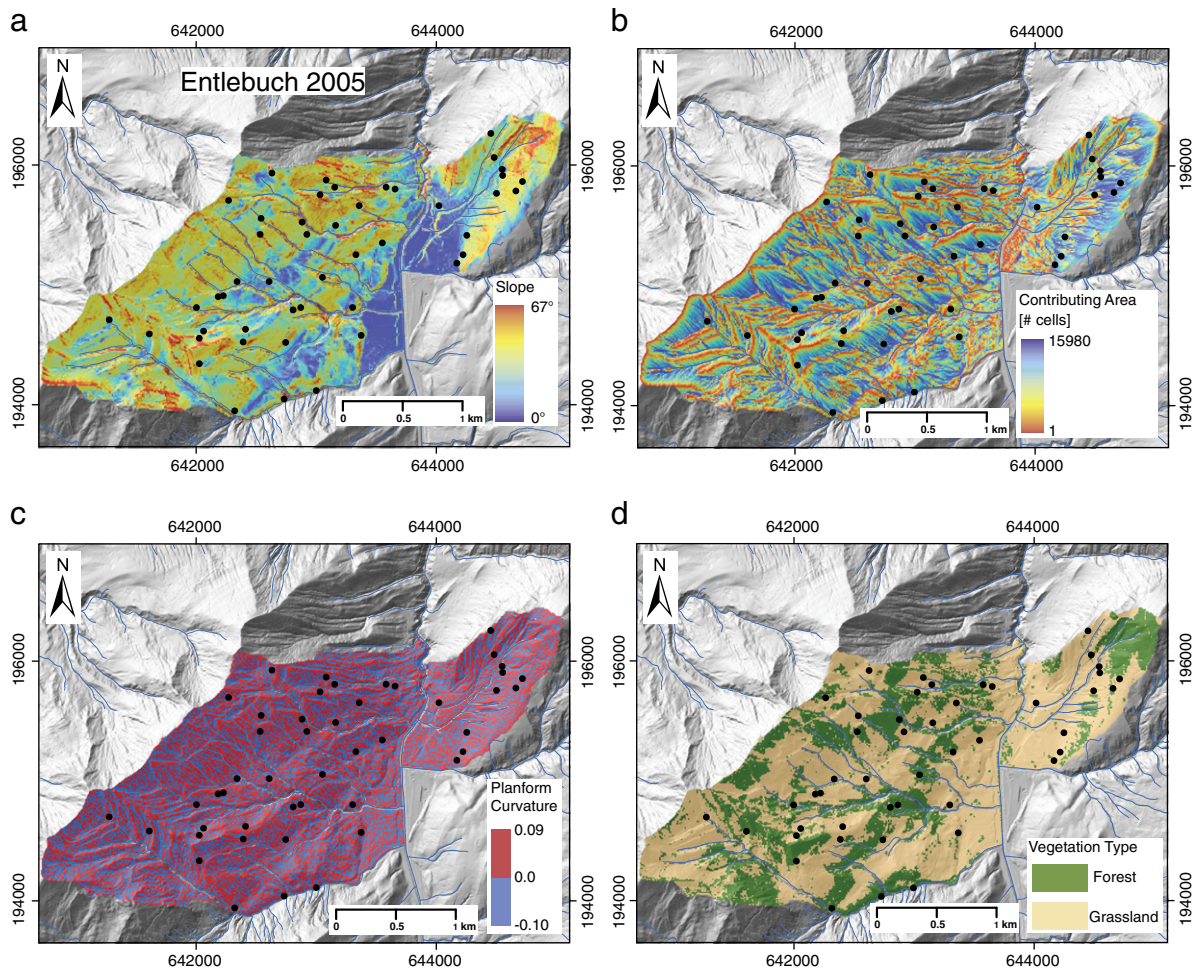


Fig. 5. Maps of explanatory variables (10 m grid spacing) used to model landslide occurrence for study area Entlebuch 2005 with locations of shallow landslides (black dots): (a) slope angle, (b) contributing area, (c) planform curvature, and (d) vegetation type.

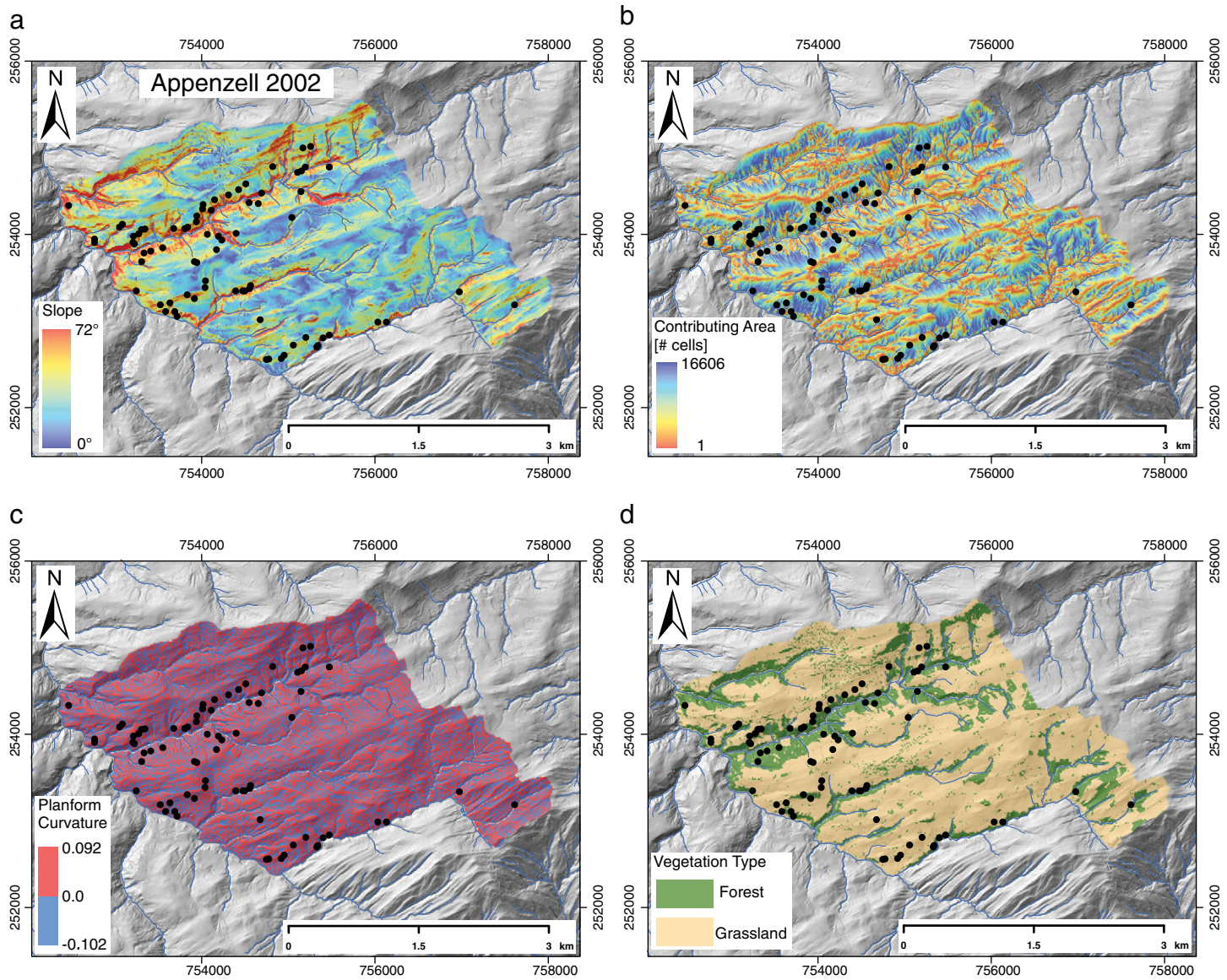


Fig. 6. Map of explanatory variables (10 m grid spacing) used to model landslide occurrence for study area Appenzell 2002 with locations of shallow landslides (black dots): (a) slope angle, (b) contributing area, (c) planform curvature, and (d) vegetation type. Steep slopes exist only along the incised valleys.

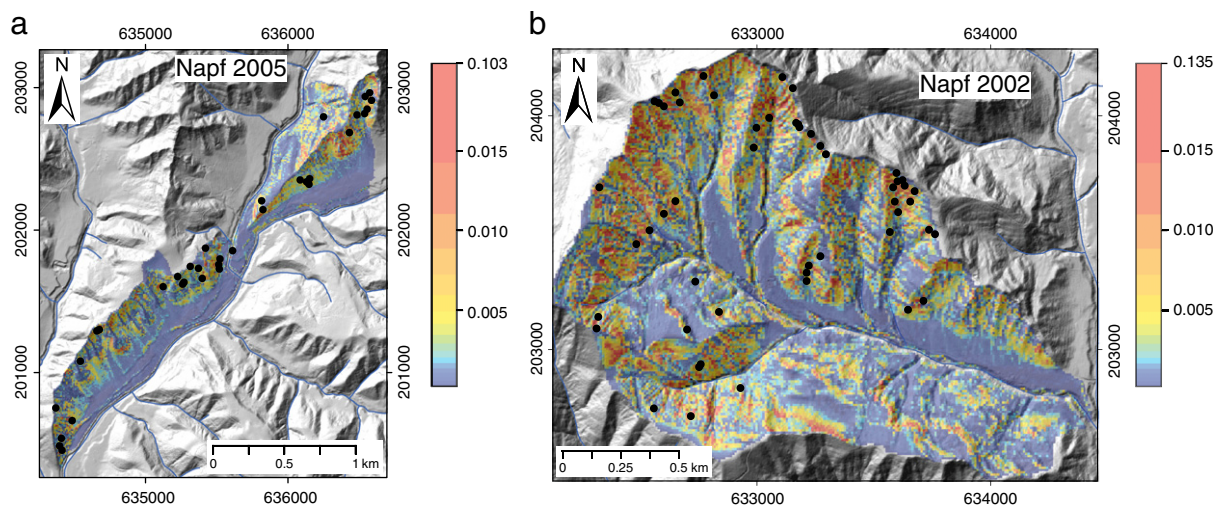


Fig. 7. Susceptibility maps (10 m grid spacing) for Napf study areas with locations of shallow landslides indicated by black dots. (a) Calibration study area Napf 2005 (34 landslides, one landslide discarded as outlier). (b) Validation study area Napf 2002, for which the sum of the predicted susceptibilities (93) exceeded the observed number (51) of landslides.

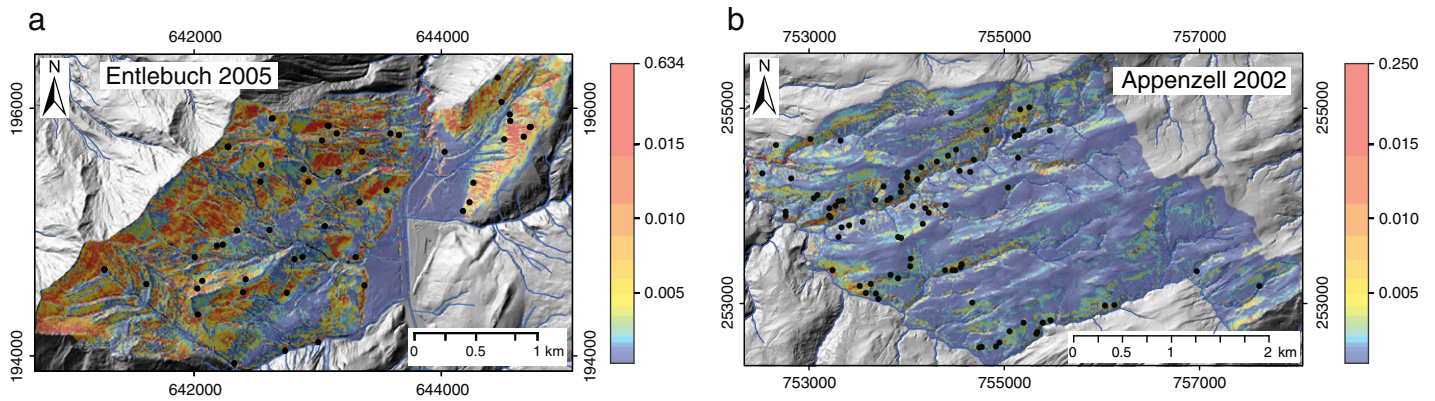


Fig. 8. Susceptibility maps (10 m grid spacing) for study areas (a) Entlebuch 2005 and (b) Appenzell 2002 with locations of shallow landslides indicated by black dots. Landslides in Entlebuch 2005 were triggered by the same rainfall as in case of Napf 2005. The sum of the predicted susceptibilities (Entlebuch 325, Appenzell 186) exceeded the observed numbers (Entlebuch 46, Appenzell 76) of landslides.

more pronounced planform curvature and more grassland. These results agree with our understanding of the triggering processes. The driving forces increase with the slope angle, and the resisting forces are weaker in the absence of tree roots. Differences in the soil water regime between forest and grassland due to contrasting interception and transpiration may also play a role. Planform curvature and contributing area are both linked to water flow paths in hillslopes: converging and amassing flow results in larger driving forces. The effect of the two variables is not fully independent, but the small Spearman correlation coefficient (0.33) suggests that each variable accounts for effects not explained by the other. The two other variables linked to water flow on hillslopes, profile curvature and distance to the nearest drainage pathway, were not selected for the final model, suggesting that the mere amount of water flowing through a cell mainly controls the driving forces of landslides.

5.2. Predictive power of the logistic regression model

Although the maps of landslide susceptibility (Figs. 7 and 8) indicate that landslides occurred in the zones with higher predicted susceptibilities, the susceptibility predictions were positively biased

for all validation areas. Furthermore, according to Hosmer and Lemeshow (2000), the quality of the predictions was just ‘acceptable’ for Napf 2002 and Entlebuch 2005 ($AUC \sim 0.7$). These two areas are of special interest because the first area is close to the calibration study area and thus has similar geomorphic processes, and in the second area the landslides have been triggered by the same rainfall as for Napf 2005. Although these two case studies led to no sound statement, the slight advantage for Napf 2002 (same geomorphology) indicates that landslides were not controlled by rainfall characteristics alone.

Interestingly, the AUC value of 0.82 for Appenzell 2002 (different geomorphology and rainfall event) was as good as the AUC value of the apparent ROC curve. This can be explained by the fact that the Appenzell study area mainly consists of extended flat areas and the steep slopes are constrained to local valleys. Since slope angle was the most influential single explanatory variable, the predicted susceptibilities were small for many grid cells (flat region) and the model thereby correctly classified these grid cells as landslide-free. Due to the clear spatial separation between flat areas without landslides and steep regions with many landslides, the model managed to discriminate between areas of high and low susceptibility.

The AUC value of the cross-validation ROC curve was 0.74 and thus not much larger than the AUC value of the validation ROC curve for Napf 2002 (0.72). Hence, unlike goodness-of-fit measures, the cross-validation statistic provides quite a realistic picture of the predictive power of the statistical model for independent study areas, at least if the calibration and validation areas are not far apart and thus geomorphic processes are similar.

6. Conclusions

We mapped landslide susceptibility in four small study areas in the northern foothills of the Swiss Alps by logistic regression, based on event-based inventories of rainfall-triggered landslides and high resolution digital elevation models. Information on geology, soil type and land use was not considered because of poor spatial resolution of available data. A model was fitted to the landslide data of one study area. The estimated regression coefficients implied that susceptibility increases with increasing slope angle, more pronounced planform curvature, absence of tree roots and increasing contributing area. The model was then validated for another nearby study area with similar geomorphology but with landslides triggered by another rainfall event. Validation was also made for a second study area with different geomorphology and geology but landslides triggered by the same rainfall event. The latter resulted in slightly worse susceptibility predictions, as expressed by the area under the ROC curve ($AUC = 0.69$ vs. 0.72). For the model fitted to the calibration data, the

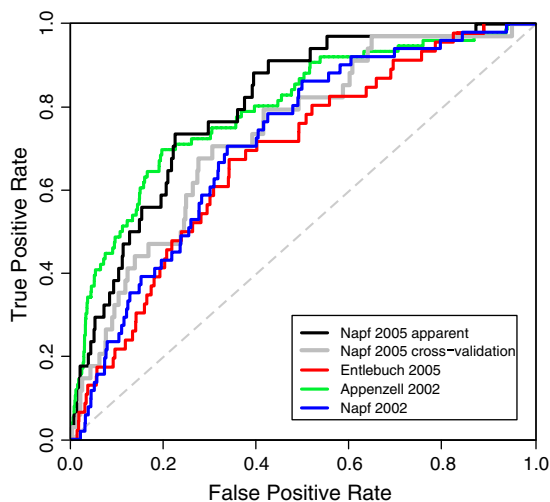


Fig. 9. Receiver Operating Characteristics (ROC) curves used to assess the predictive performance of the logistic regression model. For Napf 2005 both the apparent and the cross-validation ROC curves (20-fold cross-validation) are shown. Cross-validation AUC for Napf 2005 is 0.74 and the apparent AUC value is 0.80. For the Napf 2002 and Entlebuch 2005 validation study areas, the AUC values were somewhat smaller (0.72 and 0.69 respectively) and for Appenzell 2002 the AUC value was 0.82.

AUC value was 0.80, confirming that the goodness-of-fit criteria overrate the predictive performance of a statistical model for independent test data. Twenty-fold cross-validation provided a more realistic measure of the predictive performance ($AUC=0.74$). The good predictive performance ($AUC=0.82$) for another case with different geomorphology and landslides triggered by a different rainfall event can be explained by the large area of gentle slopes for which the regression model correctly predicted no landslides.

Although the model could delineate the zones of actual landslides reasonably well, the predictive power according to AUC values was not excellent, suggesting that terrain features and plant cover do not solely determine the occurrence of landslides. For example, the depth and hydro-mechanical properties of soil are very important for the water storage and flow that affect landslides. To improve landslide susceptibility predictions, we need detailed spatial information on soil. In addition, a predictive model based on static land properties cannot fully reflect the complex and time dependent processes of landslides. However, this study shows that such a model is still capable of distinguishing zones highly or less prone to shallow landslides. A statistical model may also point to major factors controlling landslides, and such information is useful for developing physically based models to simulate landslide triggering processes in a more realistic manner.

Acknowledgments

This work is part of the projects 'Local and regional hydrologic and geomorphic factors determining landslide patterns', funded by Swiss National Science Foundation (SNSF), and 'Triggering of Rapid Mass Movements in Steep Terrain (TRAMM)', funded by the Competence Centre Environment and Sustainability (CCES) of the ETH Domain.

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